

Quantum Entanglement, Part 3

Lecture 4: Proper Time, Time Dilation, and Length Contraction

Leonard Susskind

Transcript-derived notes curated by [LazyingArt LLC](#)

Produced with [Video2Book](#)

Chapter 1

Proper Time, Time Dilation, and Length Contraction

We shall often choose units in which the speed of light is $c = 1$. In those units a year of time and a light-year of distance may be compared directly, and the formulas are much easier to read. Whenever c matters physically, we shall put it back. Dimensional analysis is enough to locate it.

1.1 The Galilean Spacetime Grid

Begin with the pre-Einstein relation between a frame at rest and a train moving at velocity v along the x -axis:

$$x' = x - vt, \quad t' = t. \quad (1.1)$$

The second equation contains Newton's assumption of universal time. Every observer uses the same simultaneity slices, even though the spatial origins move relative to one another.

Suppose the unprimed observer stands beside the track and the primed observer rides the train. The train's origin has $x' = 0$, so its worldline in the unprimed coordinates is

$$x = vt. \quad (1.2)$$

That sloping line is the t' -axis. By contrast, $t' = t$ says that all lines of constant t' are the same horizontal lines as those of constant t .

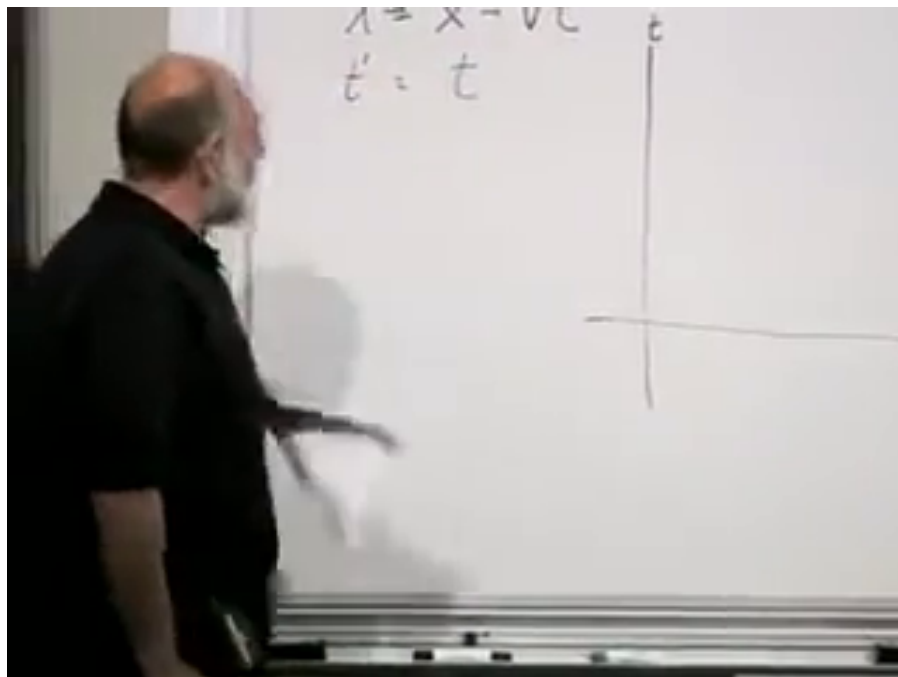


Figure 1.1: The Galilean transformation and the initial t - x grid. The moving origin will trace $x = vt$, while the two frames retain common constant-time lines.

Lecture frame: 00:03:06.

Solving Eq. (1.1) for the unprimed coordinates gives

$$x = x' + vt' = x' + vt, \quad t = t'. \quad (1.3)$$

The inverse has the same form with v replaced by $-v$. If the train moves forward relative to the station, the station moves backward relative to the train. This reciprocity survives in special relativity, but universal time does not.

1.2 The Lorentz Correction

Galilean transformations cannot make the speed of light the same in every inertial frame. The corrected relations are, first in $c = 1$ units,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \quad t' = \frac{t - vx}{\sqrt{1 - v^2}}. \quad (1.4)$$

Restoring ordinary units gives

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \quad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}. \quad (1.5)$$

$$X' = \frac{X - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$t' = \frac{t - \frac{v}{c^2}x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Figure 1.2: The Lorentz transformation with the factors of c restored.*Lecture frame: 00:09:10.*

For $v \ll c$, both v^2/c^2 and vx/c^2 are tiny in the circumstances where Newtonian mechanics is used. Equation (1.5) then reduces to the Galilean result. Relativity does not discard Newtonian kinematics; it identifies its range of validity.

At $v = c$, the denominator vanishes. An inertial frame cannot move at the speed of light relative to another inertial frame. Well below that limit the corrections are small, but near it the mixing of x and t becomes unavoidable.

Motion was chosen along x , so the transverse coordinates do not change:

$$y' = y, \quad z' = z. \quad (1.6)$$

One way to see why is to place identical transverse meter sticks on the train and platform. An observer moving halfway between the two frames sees the rods approach with equal and opposite speeds. The setup has no feature that could make one transverse rod shorter than the other.

1.3 Lorentz Transformations Along the Light Cone

There is a geometric way to read Eq. (1.4). Add its two equations:

$$t' + x' = \frac{(t - vx) + (x - vt)}{\sqrt{1 - v^2}} \quad (1.7)$$

$$= (t + x) \frac{1 - v}{\sqrt{1 - v^2}} \quad (1.8)$$

$$= (t + x) \sqrt{\frac{1 - v}{1 + v}}. \quad (1.9)$$

Subtract them in the complementary order:

$$t' - x' = \frac{(t - vx) - (x - vt)}{\sqrt{1 - v^2}} \quad (1.10)$$

$$= (t - x) \frac{1 + v}{\sqrt{1 - v^2}} \quad (1.11)$$

$$= (t - x) \sqrt{\frac{1 + v}{1 - v}}. \quad (1.12)$$

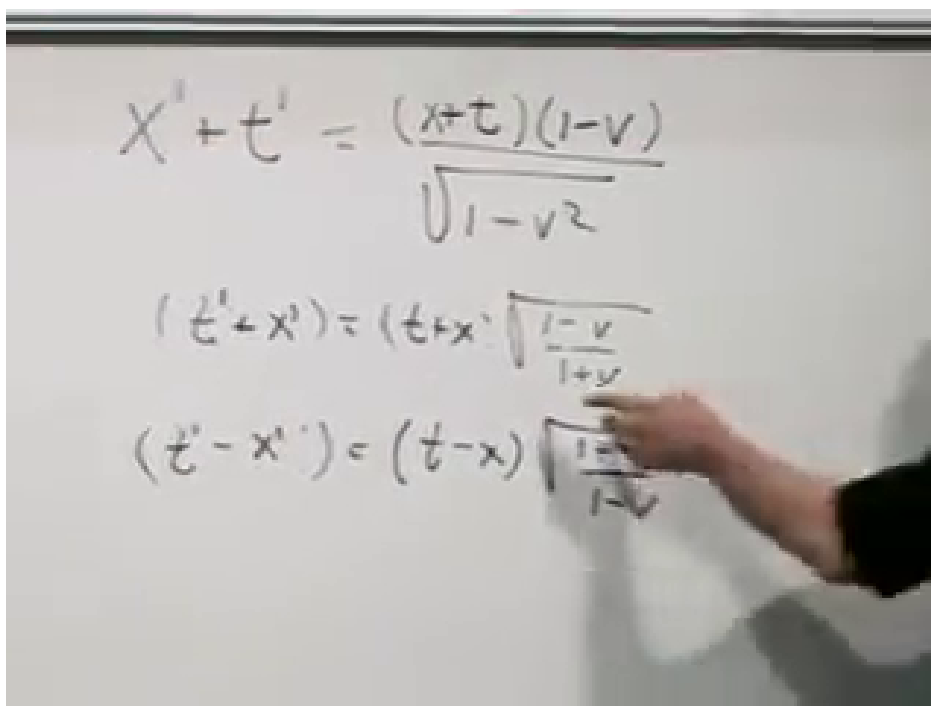


Figure 1.3: The Lorentz transformation rewritten in the null combinations $t + x$ and $t - x$. Their scale factors are reciprocal.

Lecture frame: 00:18:45.

The combinations $t + x$ and $t - x$ are coordinates along the two 45° lightlike directions. A boost squeezes one of these directions and stretches the other by the inverse factor. It does not rotate either one away from 45° . Thus light rays remain light rays, which is the spacetime picture of the invariant speed of light.

Question from the audience. The halfway observer sees a symmetric experiment, but how does that establish the result for the train and platform observers? ^a

Response. One additional fact is being used: a coincidence of events is invariant. If the ends of the rods meet, that meeting is one spacetime event. Every inertial observer may assign different coordinates to it, but none may split one coincidence into two different locations. Flashes emitted where the corresponding endpoints meet make those events operationally detectable. The symmetry of the halfway frame and the invariance of the two endpoint coincidences establish that the transverse lengths agree.

^aLecture timestamp: 00:20:48.

Question from the audience. Must the bicyclist who represents the halfway frame be located physically at the midpoint of the rods?^a

Response. No. A reference frame is a whole network of observers and clocks, not one eye at one place. The bicyclist may receive the endpoint flashes later and infer where they originated. What matters is the spacetime location of each emission event, not where the light is eventually received.

^aLecture timestamp: 00:22:25.

The composition of successive velocities is another consequence of the same transformation, but it is deliberately left for the next lecture. The present task is to identify the invariant quantity that a moving clock measures.

1.4 The Invariant Interval

For a rotation of ordinary Cartesian axes, the quantity $x^2 + y^2$ is unchanged. Lorentz transformations preserve a different quadratic form. From Eq. (1.4),

$$t'^2 - x'^2 = \frac{(t - vx)^2 - (x - vt)^2}{1 - v^2} \quad (1.13)$$

$$= \frac{t^2 - 2vtx + v^2x^2 - x^2 + 2vtx - v^2t^2}{1 - v^2} \quad (1.14)$$

$$= t^2 - x^2. \quad (1.15)$$

The mixed terms cancel, and the remaining numerator contains a common factor of $1 - v^2$.

The whiteboard shows the following handwritten derivations:

$$x' = \frac{x - vt}{1 - v^2}$$

$$t' = \frac{t - vx}{1 - v^2}$$

$$t'^2 - x'^2 = \frac{(t - vx)^2 - (x - vt)^2}{1 - v^2}$$

$$= \frac{t^2 - 2vtx + v^2x^2 - x^2 + 2vtx - v^2t^2}{1 - v^2}$$

$$= \frac{t^2(1 - v^2) - x^2(1 - v^2)}{1 - v^2}$$

$$= t^2 - x^2$$

Figure 1.4: The board derivation of $t'^2 - x'^2 = t^2 - x^2$.

Lecture frame: 00:33:15.

With three spatial dimensions the invariant is

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2 \quad (1.16)$$

in $c = 1$ units. More generally, for two events we use coordinate differences:

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \quad (1.17)$$

Every inertial observer agrees on this number even though they disagree on its separate time and space terms.

1.5 Proper Time and Moving Clocks

Imagine a lattice of stationary observers whose clocks have been synchronized by exchanging light signals. Their readings define the coordinate time t . Now carry one wristwatch through that lattice. Its own reading follows the watch along its worldline; that reading is proper time.

For two timelike-separated events, define

$$c^2 \Delta \tau^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \quad (1.18)$$

In one spatial dimension with $c = 1$, this is simply

$$\Delta \tau = \sqrt{\Delta t^2 - \Delta x^2}. \quad (1.19)$$

The square root parallels the formula for an ordinary distance, but the minus sign changes the geometry.

There are three cases:

$$\Delta t^2 - \Delta x^2 > 0 \quad \text{timelike,} \quad (1.20)$$

$$\Delta t^2 - \Delta x^2 = 0 \quad \text{lightlike,} \quad (1.21)$$

$$\Delta t^2 - \Delta x^2 < 0 \quad \text{spacelike.} \quad (1.22)$$

Inside the light cone the proper time is real. On the cone, where $x = \pm t$, it is zero even though the events need not coincide. A light ray can connect precisely such events. Outside the cone the separation is spacelike; it is better to describe it by the real proper distance $\sqrt{\Delta x^2 - \Delta t^2}$ than by an imaginary clock time.

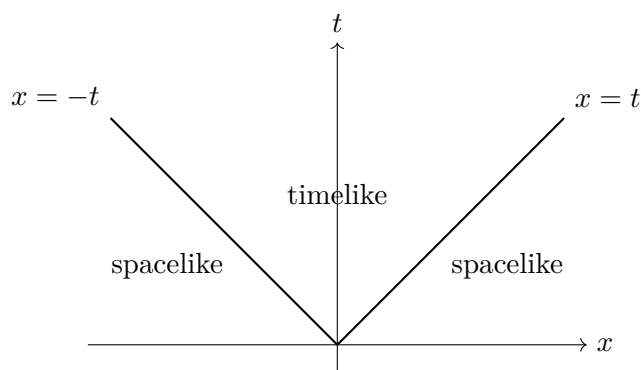
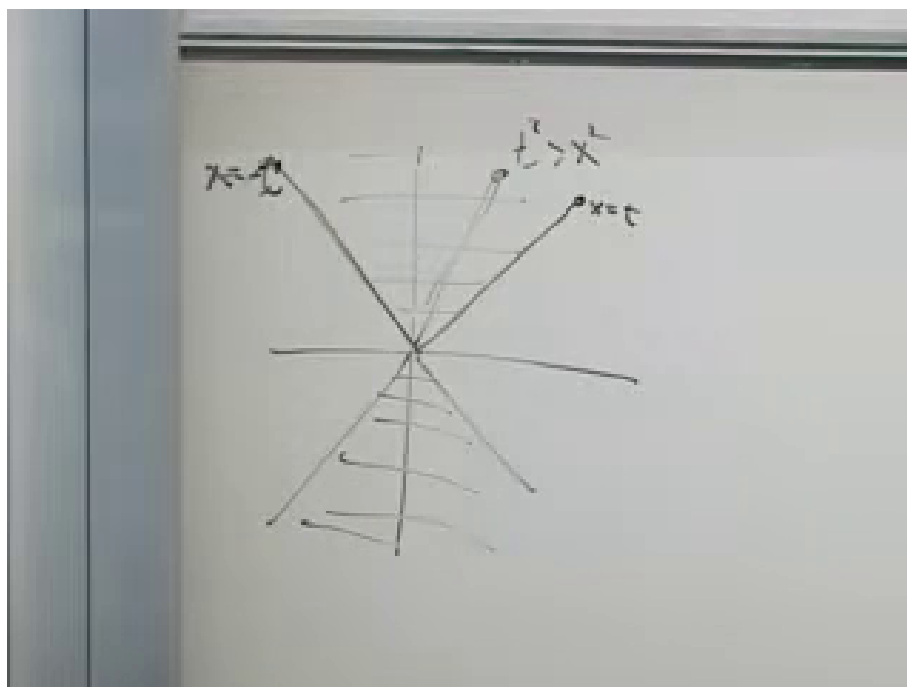


Figure 1.5: The board light cone and a clean classification of timelike, lightlike, and spacelike separation.

Lecture frame: 00:38:40.

Why does Eq. (1.18) equal a clock reading? For a clock at rest, $\Delta x = 0$, so $\Delta\tau = \Delta t$. For a uniformly moving clock, choose the primed frame in which that clock is at rest. There $\Delta x' = 0$, and invariance gives

$$\Delta\tau = \sqrt{\Delta t'^2 - \Delta x'^2} = \Delta t'. \quad (1.23)$$

Moving clocks therefore tick off proper time.

Question from the audience. Are “the interval between the events” and “the reading of the clock” being used as two names for the same thing? ^a

Response. They coincide when an ideal clock travels inertially from one event to the other. The invariant interval is a geometric quantity; the difference between the endpoint readings of the comoving clock is its physical measurement. A stationary coordinate clock generally records a different coordinate-time interval.

^aLecture timestamp: 00:44:49.

1.6 Time Dilation and Simultaneity

Consider the origin of the moving frame. Its worldline obeys

$$x' = 0, \quad x = vt. \quad (1.24)$$

The same sloping line is both the path of the moving clock and the t' -axis.

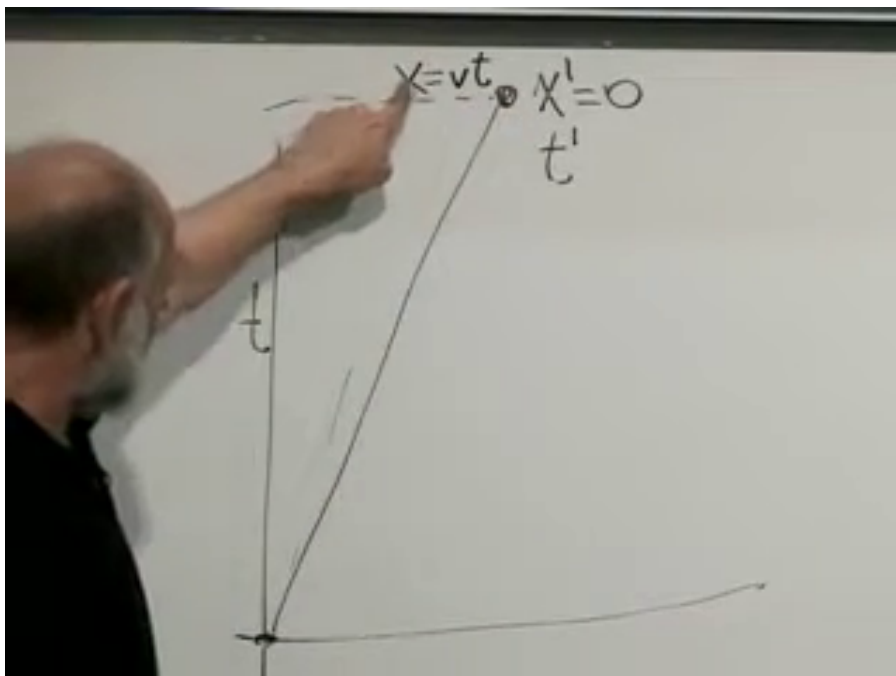


Figure 1.6: The moving origin: $x = vt$, $x' = 0$, and the t' -axis are three descriptions of the same worldline.

Lecture frame: 00:48:18.

Apply the invariant interval between the common origin and a later point on that worldline:

$$t'^2 - x'^2 = t^2 - x^2, \quad (1.25)$$

$$t'^2 = t^2 - v^2 t^2, \quad (1.26)$$

$$t' = t\sqrt{1 - v^2}. \quad (1.27)$$

With c restored,

$$\Delta\tau = \Delta t \sqrt{1 - \frac{v^2}{c^2}} = \frac{\Delta t}{\gamma}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (1.28)$$

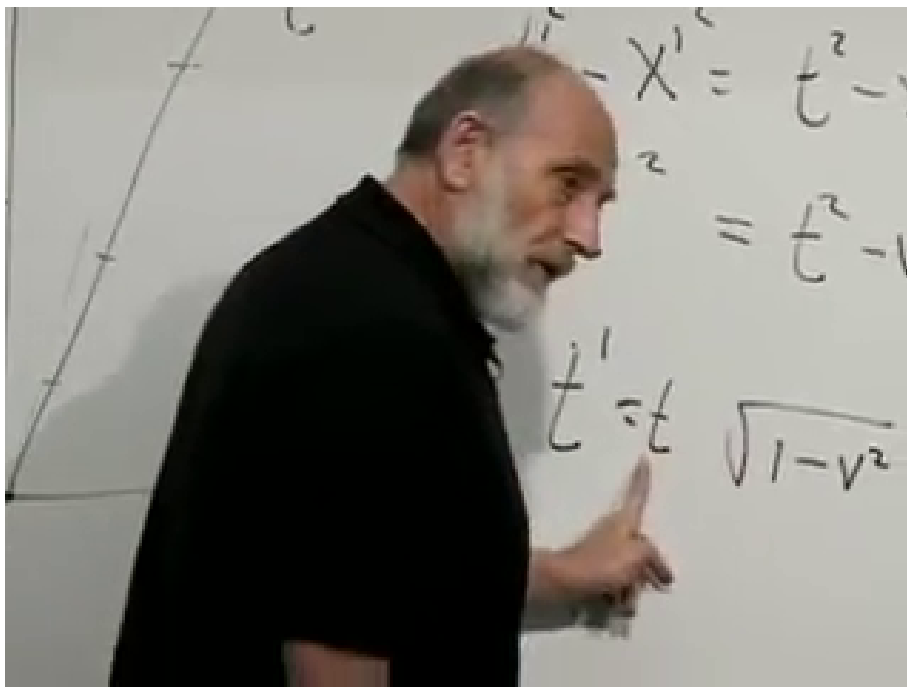


Figure 1.7: The time-dilation result on the board: the moving clock accumulates $t' = t\sqrt{1 - v^2}$ in units with $c = 1$.

Lecture frame: 00:50:45.

The stationary observer says that the moving clock runs slow. Reciprocity requires the moving observer to say the same about stationary clocks. There is no contradiction because the two descriptions compare different pairs of distant events.

Indeed, a constant value of t' in Eq. (1.4) satisfies

$$t - vx = \text{constant}. \quad (1.29)$$

The moving observer's simultaneity lines are tilted relative to the horizontal lines $t = \text{constant}$. Each observer calls a different collection of distant clock readings simultaneous. Once that difference is kept visible, the reciprocal slowing of clocks is consistent.

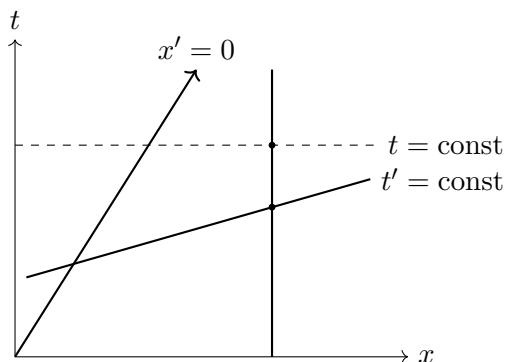


Figure 1.8: Reciprocal time dilation compares clock readings on different simultaneity slices.

1.7 The Twin Paths

Two twins begin at the same event. One remains at $x = 0$. The other travels outward at speed v , reverses direction, and returns at the same speed. When they reunite they compare every physical process that can serve as a clock: watches, heartbeats, metabolism, or aging itself.

Let each half of the trip take coordinate time T in the home frame. The home twin's proper time is

$$\tau_{\text{home}} = 2T. \quad (1.30)$$

The turn-around event has coordinates (vT, T) . The first traveling leg therefore contributes

$$\tau_{\text{leg}} = \sqrt{T^2 - v^2 T^2} = T\sqrt{1 - v^2}. \quad (1.31)$$

The return leg contributes the same amount, hence

$$\tau_{\text{travel}} = 2T\sqrt{1 - v^2} < 2T. \quad (1.32)$$

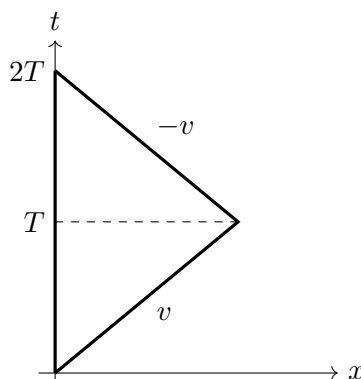
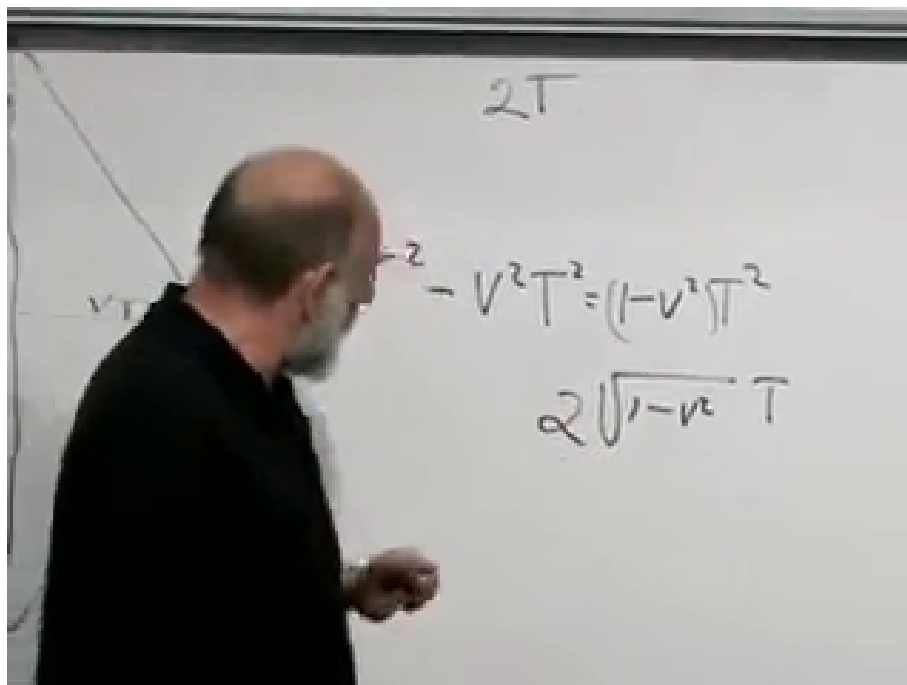


Figure 1.9: The home worldline and the two traveling legs. Their proper times are different because they are different paths through spacetime.

Lecture frame: 00:59:35.

There is no contradiction in the unequal ages. The traveling path contains a turn and is not related to the straight home path by a single inertial-frame symmetry. In ordinary Euclidean geometry a straight route and a broken route between two points also have different lengths. Minkowski geometry reverses which path has the larger interval, but the comparison is still geometric.

1.8 Acceleration and the Ideal Clock

Question from the audience. At the turn-around the physics is different and the traveler accelerates. Does that invalidate the clock calculation? ^a

Response. The acceleration is real and helps identify the asymmetry; it should not be argued away. The sharp corner is only an idealized drawing and may be rounded into a finite turn. The further physical assumption is that a sufficiently small, well-built clock is insensitive to modest acceleration and continues to record proper time. A large or fragile clock may deform or break, but that is a defect of the mechanism rather than of the spacetime interval.

^aLecture timestamp: 01:02:19.

This ideal-clock assumption is not derived from the Lorentz transformation alone. It is tested through the behavior of actual clock mechanisms. An atom is a much better approximation than a clock hundreds of miles across because the compact mechanism is harder to distort under acceleration or gravity. Invoking general relativity by name would not eliminate this physical issue; one would still have to know how the chosen mechanism responds to its environment.

Question from the audience. Can an accelerated trajectory be reduced to short intervals on which the inertial analysis remains valid? ^a

Response. Yes. Divide the curve into short, nearly straight segments, calculate the proper time on each segment, and add the results. This is the spacetime analogue of finding the length of an ordinary curve from many small line segments. In the continuum limit,

$$\tau = \int \sqrt{1 - \frac{v(t)^2}{c^2}} dt,$$

for motion in flat spacetime described in an inertial coordinate system.

^aLecture timestamp: 01:09:08.

1.9 What the Twins See

The final clock readings can be made operational by sending light pulses. Mark one-second intervals of proper time on both worldlines and let each twin emit a pulse at every mark. During the outward leg, pulses received from the other twin are spread apart. During the return, they arrive bunched together. This is the relativistic Doppler effect expressed directly on the spacetime diagram.

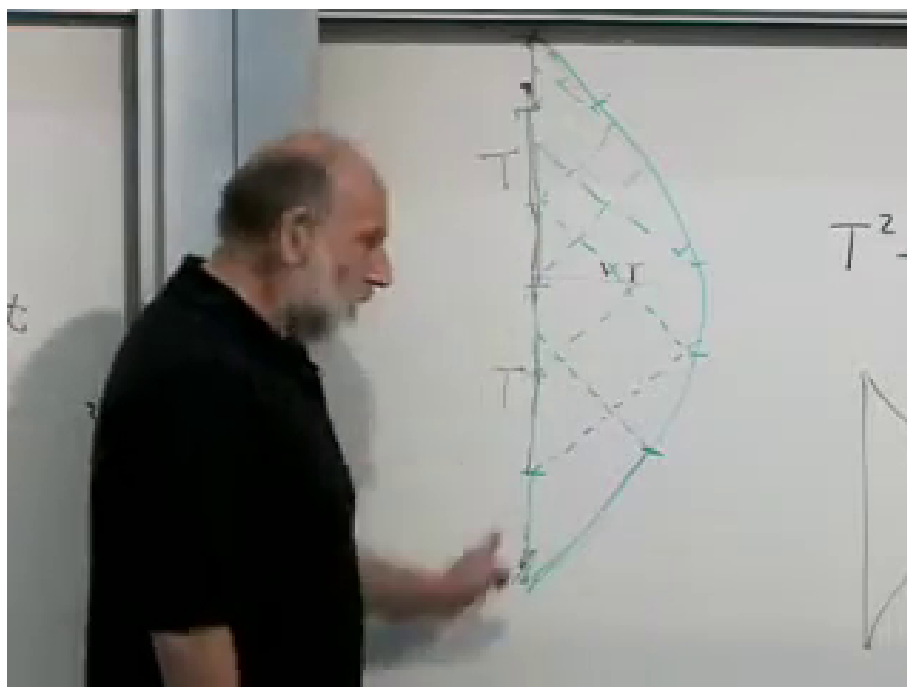


Figure 1.10: Light signals exchanged between the home and traveling worldlines. Sparse arrivals on the outward leg and compressed arrivals on the return leg account for every emitted pulse.

Lecture frame: 01:12:55.

No frames of the story disappear. Each observer eventually receives all the pulses emitted by the other, but their reception rates depend on the relative motion. The outbound redshift and inbound blueshift are not equal-duration effects for the two paths, so counting all received pulses agrees with the proper-time comparison.

Question from the audience. If the returning traveler receives blueshifted light, could the traveler collect more energy than the source emitted? ^a

Response. That question requires the relativistic definition and frame dependence of energy, which have not yet been developed. It is reasonable, but coordinates and pulse counts alone are not enough to settle the energy bookkeeping, especially for an accelerated observer. The answer should therefore be postponed rather than guessed.

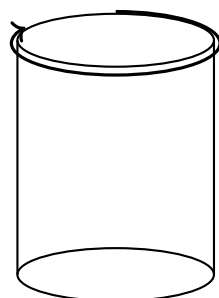
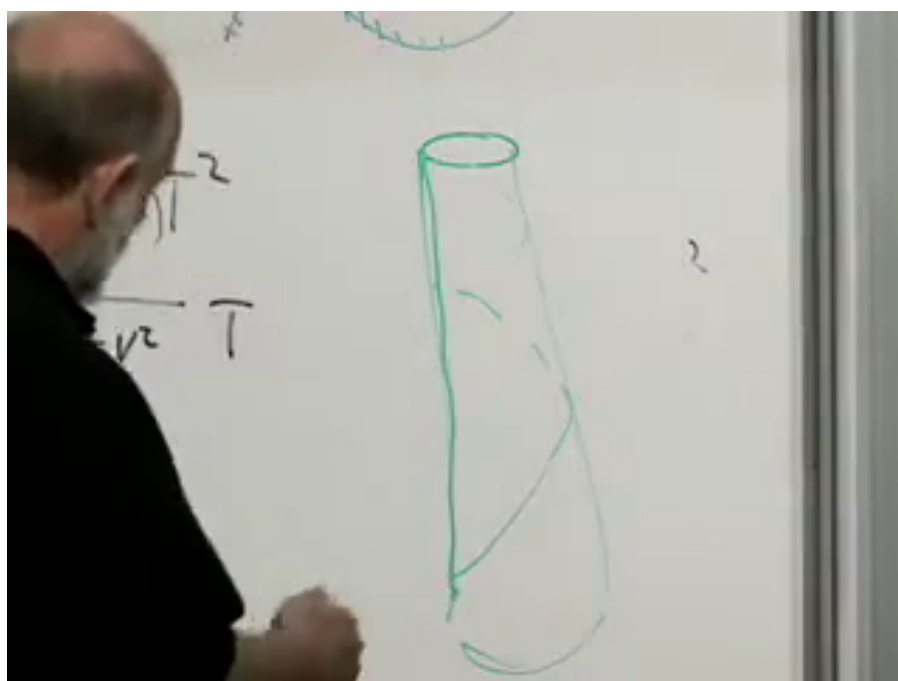
^aLecture timestamp: 01:13:50.

1.10 A Closed Spatial Direction

Question from the audience. Suppose one spatial direction is compact. Could a twin move uniformly around it, return to the starting place, and avoid acceleration altogether? What then distinguishes the twins?^a

Response. The space should be imagined intrinsically as a cylinder, not as a sheet bent inside a larger space; there need be no radial acceleration. The returning observer nevertheless encounters the global fact that the same places recur. That indicates an asymmetry, but a complete answer requires a careful treatment of special relativity with the spatial identification. The question is left open here rather than answered improvisationally.

^aLecture timestamp: 01:15:14.



compact spatial direction

Figure 1.11: The compact-space question as drawn on the board, with a clean intrinsic cylinder. It is a question about global identification, not motion around a cylinder embedded in an extra radial dimension.

Lecture frame: 01:17:56.

1.11 Length Is Measured on a Simultaneity Slice

For a special-relativity problem, first draw the spacetime picture. Take a meter stick at rest in the unprimed frame, with endpoints at $x = 0$ and $x = 1$. As time passes, the two endpoints trace vertical worldlines. The stick sweeps out a ribbon between them.

The moving observer must measure both endpoints at one value of the moving time t' . From Eq. (1.29), the slice through the origin with $t' = 0$ satisfies

$$t = vx \quad (1.33)$$

in $c = 1$ units. It intersects the right endpoint $x = 1$ at $t = v$.

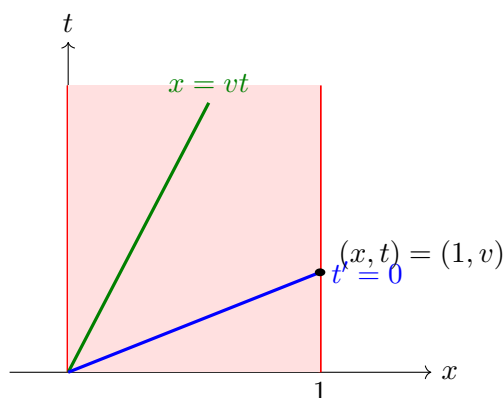
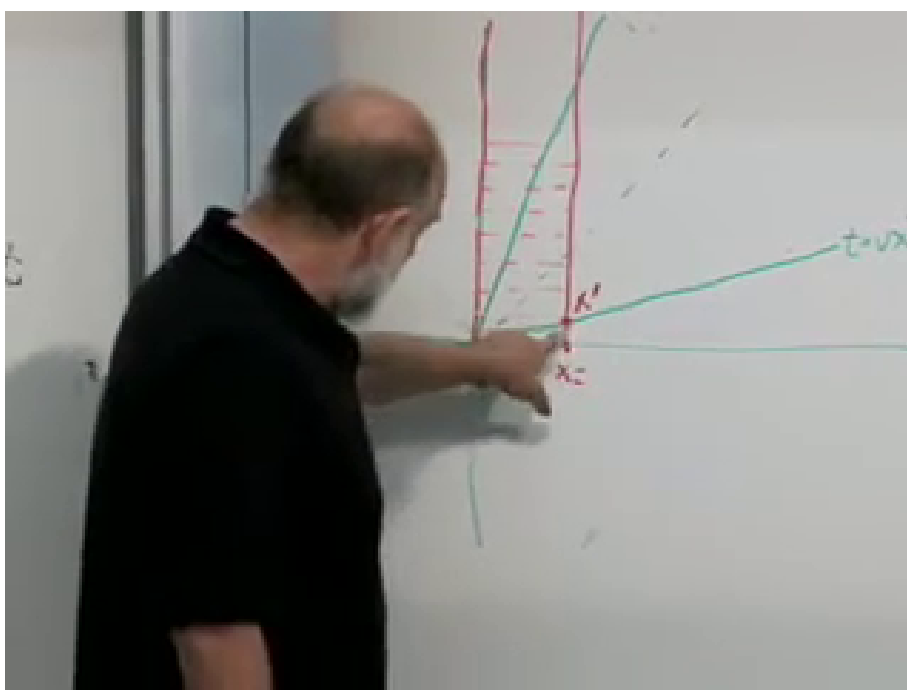


Figure 1.12: A rod at rest sweeps out a spacetime ribbon. The moving observer's length is the separation between its endpoints on one $t' = \text{const}$ slice.

Lecture frame: 01:24:20.

The selected endpoint separation is spacelike, so use the positive spatial form of the invariant:

$$x'^2 - t'^2 = x^2 - t^2. \quad (1.34)$$

At the chosen endpoint, $t' = 0$, $x = 1$, and $t = v$. Hence

$$x'^2 = 1 - v^2, \quad (1.35)$$

$$x' = \sqrt{1 - v^2}. \quad (1.36)$$

Thus a rest length L is measured in the moving frame as

$$L' = L\sqrt{1 - v^2/c^2} = \frac{L}{\gamma}. \quad (1.37)$$

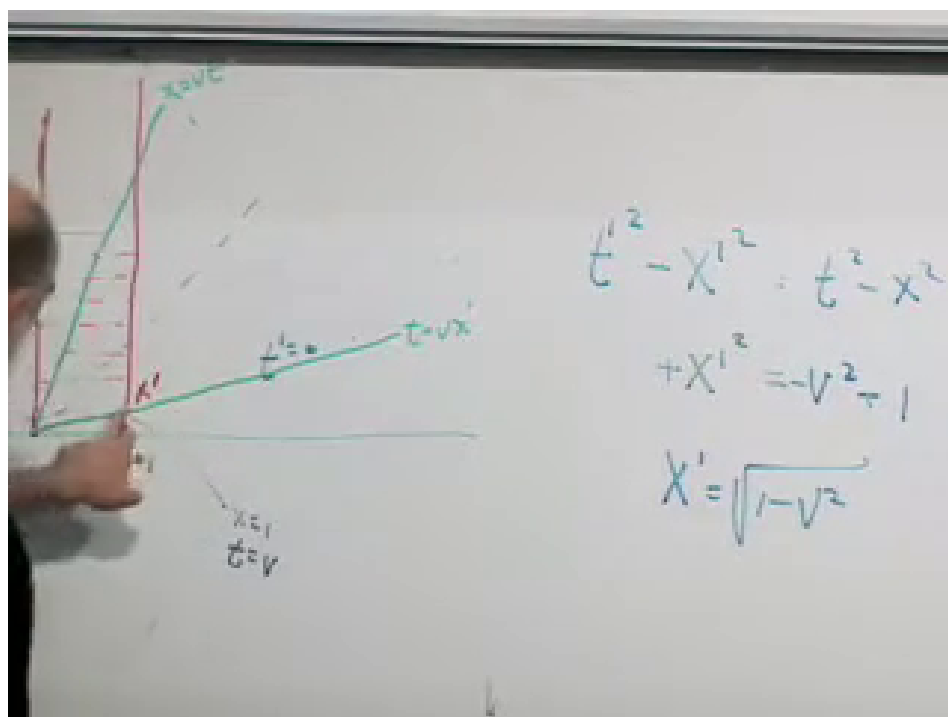


Figure 1.13: The completed length-contraction construction and the board result $x' = \sqrt{1 - v^2}$ for a unit rest length.

Lecture frame: 01:28:10.

The reciprocal statement is also true: the platform observer assigns a shorter length to the train's longitudinal meter stick. Once again the two observers choose different simultaneous endpoint events, so reciprocity is not contradictory.

1.12 Questions at the End

Question from the audience. Does anything besides light propagate at the speed of light? In particular, what about gravity?^a

Response. General relativity predicts gravitational waves that propagate at c . At the time of this lecture, the evidence came indirectly from systems such as binary pulsars; gravitational waves had not yet been detected directly.^b

^aLecture timestamp: 01:29:29.

^bEditorial clarification: The experimental situation changed after this 2006 lecture. LIGO directly detected GW150914 in 2015, and the joint observation of GW170817 and its electromagnetic counterpart in 2017 showed that the propagation speeds of gravity and light agree to within a few parts in 10^{15} . See <https://ligo.org/detections/gw150914/> and <https://doi.org/10.1103/PhysRevLett.119.251302>.

Question from the audience. Can an astronaut cross a distance of many light-years in less proper time than the distance expressed in light-years? Would that mean moving faster than light?^a

Response. The astronaut's proper time is $\sqrt{\Delta t^2 - \Delta x^2}$, which is smaller than the coordinate time Δt . In the astronaut's inertial frame, the distance between the stars is Lorentz-contracted. The stars move with speed $-v$, not faster than light; there is simply less contracted distance to cross.

^aLecture timestamp: 01:30:38.

If the astronaut begins at rest and accelerates, the description changes between a succession of instantaneous inertial frames. The distance assigned to the stars can change during that transition. One cannot apply one fixed Lorentz transformation to the entire accelerated history, but the total proper time still follows by adding the contributions along the worldline. The same statement applies to a realistic twin who launches, turns, and decelerates rather than being born at cruising speed.

Question from the audience. Are these puzzles caused by treating time as special? Must we call it a fourth dimension?^a

Response. It is not necessary to begin with the slogan that time is a fourth dimension. The Lorentz transformation itself forces space and time to mix, while the interval $t^2 - x^2 - y^2 - z^2$ treats them as parts of one geometry with a different sign for time and space. At low velocity the mixing becomes negligible and familiar Newtonian intuition returns. At relativistic velocity, however, space and time cannot be regarded as fundamentally decoupled.

^aLecture timestamp: 01:36:43.

Time dilation, length contraction, the twin comparison, and shortened travel time are therefore not separate tricks. They all follow from three habits: identify the events being compared, draw their spacetime geometry, and use the invariant interval.